



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	05
Areas covered	Encapsulation

Question No 01.

Create a Java class named Person with attributes firstName and lastName. Include a constructor to initialize these attributes and a method to print the full name.

Question No 02.

Write a Java class named Teacher with attributes name and subject. Include methods to set and get the teacher's details.

Question No 03.

Create a Java class named BankAccount with the following private attributes:

- accountNumber
- balance

Include:

- a constructor to initialize the attributes
- getter and setter methods

In the setBalance() method:

- only allow positive balance values
- display "Invalid Balance" if the value is negative

Create a test class to:

- create one object
- assign values using setter methods
- display the account details using getter methods

Question No 04.

Develop a Java code base to represent a cybersecurity login system.

Create a class named User with the following private attributes:

- username
- password

Include:

- constructors
- getter and setter methods

In the setPassword() method:

- password length must be at least 8 characters
- otherwise display "Weak Password"

Create a test class to:

- create at least one user object
- assign values using methods
- display user details using getter methods

Question No 05.

Create a Java class named Student with the following private attributes:

- studentName
- studentId
- course
- GPA

Include:

- constructors
- getter and setter methods

In the `setGPA()` method:

- GPA should only accept values between 0 and 4

Create a method named `displayDetails()` to display all student details.

Create a test class to:

- create at least one student object
- assign values using methods
- display the student details